A Novel MMD-Critic-XAI-Based Intrusion Detection System for Cyber-Physical Microgrids

Gurmaan S. Panesar, *Student Member, IEEE*, Matthew Oinonen, *Member, IEEE*, and Walid G. Morsi, *Senior Member, IEEE*

***Abstract*—Cyber-physical microgrids are the target of attacks due to their reliance on internet-based automation and digital controlling devices that are not designed with built-in security. As these microgrids rely heavily on the distributed energy resources (DERs) within their systems, such cyberattacks can have consequences of high impact to such critical systems leading to power outages, equipment damage and data theft. Traditional intrusion detection systems (IDSs) used machine learning models to detect and classify such attacks; however, such models are black-boxes lacking interpretability and thus they become untrustworthy. This paper presents a novel trustworthy IDS for detecting and classifying attacks alongside disturbances in microgrids using an example-based Explainable Artificial Intelligence (XAI), namely Maximal Mean Discrepancy (MMD)-Critic, to provide trustworthy intrusion decisions. To enhance the time-frequency resolution for best feature representations in scalograms, MMD-Critic is utilized to identify the optimal voice parameters in the Continuous Wavelet Transform (CWT), enabling the Convolutional Neural Network (CNN) to accurately classify each class. The proposed IDS has been tested in real-time using OPAL-RT and IEC61850 compliant Siemens-SIPROTEC Intelligent Electronic Device (IED) to generate real Generic Object-Oriented Substation Events (GOOSE) messages. The results have shown that the use of MMD-Critic increases the intrusion detection accuracy to 98.85% by identifying individual cases where the CNN misclassified. Finally, a user interface is developed to allow grid operators to understand the model's decisions.**

***Index Terms*—Explainable AI, intrusion detection, wavelets.**

I. Introduction

CYBER-PHYSICAL microgrids rely heavily on integrating DERs internally with internet-based automation and digital controlling devices. Since such cyber-physical microgrid systems are not designed with built-in security, the increased integration of such DERs, which are often unsecured endpoints, makes such systems subject to high-impact targets causing power outages, equipment damage and data theft [1]. In U.S., Denial of Service attacks (DoS) have targeted such systems on solar/wind farms, temporarily interrupting communication between the operators and facilities. This attack emphasizes that microgrids are vulnerable to cyber-attacks and if IDS are not developed for microgrids it can lead to significant problems. IEDs in microgrids frequently employ IEC 61850 communication standard for GOOSE messages, but because of early emphasis on protocol performance over cybersecurity, they also expose the system to cyberattacks that can manipulate protection and control messages. Common attacks consist of false data injection (FDI), DoS, replay, insider threats, and message suppression (MS) [2]. With the rapid growth of microgrids, which is expected to reach $40 billion in market value by 2028 [3], there is an urgent need to investigate state-of-the-art microgrid IDS strategies. Strategies in literature emphasize classification, but equally important is the interpretability of the decisions of machine/deep learning models, which are often referred to as black-box models. In literature, existing IDS use feature-based XAI models for interpretability, consisting of Local Interpretable Model-Agnostic Explanations (LIME) and Shapley Additive exPlanations (SHAP) [4]. These explanations can make it difficult for humans to comprehend how the highlighted regions influenced the final decision [4], and if they are trustworthy explanations. Example-based XAI can provide better and understandable explanations than feature-based XAI by providing typical references from the training data to explain the decision [4]. The example-based XAI will provide explanations that will increase the operator's trust in the IDS and thus leads the way to improved adoption of AI in IDS in this context. Thus, there is a need to develop an example-based interpre table IDS approach that can detect and classify disturbances and attacks to ensure appropriate corrective measures are implemented.

A. Previous Work

Table I lists existing work in literature grouped as:

1) Non-Machine Learning Methods: Jadidi et al. in [5] and Wu et al. in [6] introduce an event-triggered IDS to detect both attacks and faults in microgrids. However, the approach has not been tested with hardware-in-the-loop (HIL), and thus the approach cannot be validated, which limits its real-world practical applicability. The work in [7] and [8] proposed attack detection in microgrids, however these methods could not sperate attacks from disturbances, which increases the false positive

Received 10 July 2026; accepted 31 August 2026. Date of publication 7 September 2026; date of current version 7 October 2026. This work was supported by the Natural Sciences and Engineering Research Council of Canada (NSERC) under Grant RGPIN-2024-05074. *(Corresponding author: Walid G. Morsi.)*

The authors are with the Electrical, Computer and Software Engineering Department, Faculty of Engineering and Applied Science, Ontario Tech University (UOIT), Oshawa, ON L1G 0C5, Canada (e-mail: gurmaan.panesar@ontariotechu.net; matthew.oinonen@ ontariotechu.net; walidmorsi.ibrahim@ontariotechu.ca).

Digital Object Identifier 10.1109/TICPS.2026.3731921

2832-7004 © 2026 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission. See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: North South University. Downloaded on October 30,2026 at 17:12:18 UTC from IEEE Xplore. Restrictions apply.

TABLE I
RECENT LITERATURE IN IDS IN CYBER-PHYSICAL MICROGRIDS

Papers	Attacks and Faults	HIL	XAI User Interface	ML Improved Utilizing XAI	Example-Based XAI
[2]	✓	✓	×	×	×
[4]	×	×	×	×	✓
[5]	✓	×	×	×	×
[6]	✓	×	×	×	×
[7]	×	✓	×	×	×
[8]	×	✓	×	×	×
[9]	✓	✓	×	×	×
[10]	✓	✓	×	×	×
[11]	×	×	×	×	×
[12]	×	✓	×	×	×
[13]	×	×	×	×	×
[14]	×	✓	×	×	×
[15]	×	✓	×	×	×
[16]	×	×	×	×	×
[17]	✓	×	×	×	×
[18]	✓	✓	×	×	×
[19]	×	×	×	×	×
[20]	×	×	×	✓	×
[22]	×	×	×	×	×
[23]	×	×	×	×	×
This work	✓	✓	✓	✓	✓

rates. In [9], a state estimation technique is used but the method was susceptible to estimation errors. The work in [10] introduces an approach to detect disturbances and DoS attacks. However, other attacks resembling disturbances such as FDI were not considered, thus leading to misclassification.

2) Machine Learning (ML) and Deep Learning (DL) Models: Regarding learning-based methods, Zhang et al. in [11] introduced a DoS attack detection method however, the method does not consider other anomalies like MS attacks, which can result in interrupted communications, reducing the reliability of the classifications. Wan et al., in [12] proposed a data driven approach to detect FDI attacks using a pattern recognition network (PRN); subsequently the PRN employs only a single hidden layer, which limits its ability to model complex data. The work in [13] introduced a CNN and long-term short memory (LTSM) model to detect attacks. To detect faults, the work in [14] employs a discrete wavelet transform (DWT) in conjunction with a support vector machine (SVM) and CNN; nonetheless, their accuracy of 100% indicates overfitting hence the model struggles on unseen data. The approach in [15] detects DoS, FDI, and latency attacks using DL; yet the work disregards disturbances that may result in variations similar to those caused by FDI attacks, potentially reducing classification accuracy. In [16], logistic regression method is used to detect attacks. However, since logistic regression is limited to binary predictions, the approach cannot be directly expanded to detect attacks alongside disturbances or other attacks. Bitirgen et al. in [17] uses ML models to detect attacks however, different attack types could not be separated. The work in [18] proposes a Network-based approach but the approach requires 5000 epochs to train making it very computationally heavy. These studies lack interpretability, which makes their models untrustworthy, as DL models are considered black boxes.

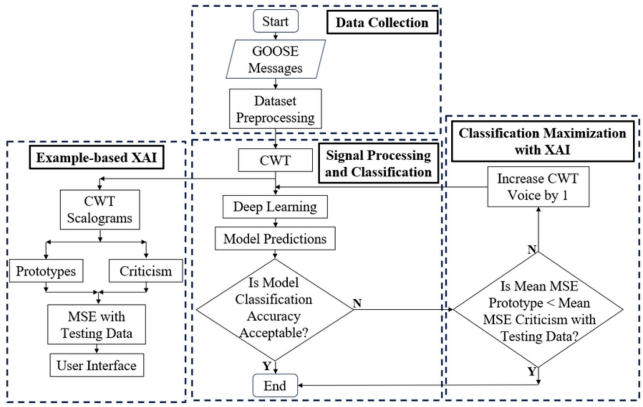


Fig. 1. Flowchart of the proposed MMD-critic-XAI-Based IDS.

3) Explainable AI (XAI) Methods: According to [19], XAI has seen limited implementation in cyber-physical systems. Nassar et al. in [2] introduces an erosion and dilation to CWT whilst using Occlusion Sensitivity Map (OSM) for explainability to detect attacks and disturbances. The work in [20] proposed combinations of CNNs and different XAI methods that give score for individual disturbances but does not analyze which features are most influential in the model's decisions. Dehbozorgi et al. in [21], proposed SVM ensemble detection method with CNN and uses interpretability with Grad-CAM to detect FDI attacks. However, each CNN in the ensemble requires its own interpretability analysis, which may lead to conflicting explanations when viewed together. The works in [22] and [23] deploy ML algorithms with SHAP and LIME for interpretability to detect DoS attacks, nevertheless, the method does not use temporal data. The main limitation of these methods is they could not use XAI to further improve the CNN decisions leading to reduced operator's trust.

B. Contributions

- The development of a novel MMD-Critic-XAI-Based IDS to provide informed and interpretable classification decisions of attacks and disturbances on a microgrid based on example-based explanations of decisions.
- MMD-Critic criticisms are leveraged to assist the CNN in better classifying difficult test cases by retraining the model with adding the criticisms that identified misclassifications back into the training data with a more detailed feature extraction process using the CWT.
- MMD-Critic is used to increase the operator interpretability of decisions through easier-to-understand example-based explanations. This is further improved through the design of a Graphical User Interface (GUI) that can be easily understood by operators at Human Machine Interface (HMI) level, thus accelerating the adoption of XAI-based IDS.

II. PROPOSED MMD-CRITIC-XAI-BASED IDS

Fig. 1 depicts the flowchart of the proposed IDS. The process begins with IEC61850 GOOSE message data packets collected

Authorized licensed use limited to: North South University. Downloaded on October 30,2026 at 17:12:18 UTC from IEEE Xplore. Restrictions apply.

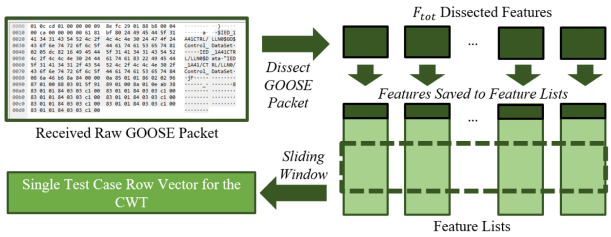


Fig. 2. A visualization of the feature preprocessing used in this work.

TABLE II
CORE FEATURES INCLUDED IN THE DATASET

Feature Name	Feature Type	Description
GOOSE_StNum	Cyber	Status number of GOOSE messages
GOOSE_SqNum	Cyber	Sequence number of GOOSE messages
GOOSE_goID	Cyber	Name of IED GOOSE messages
GOOSE_Heartbeat	Cyber	Time difference between messages
GOOSE_Boolean	Cyber	Boolean value of IEDs status

from the network, which are then processed through a preprocessing stage. These data are then used to produce time-frequency scalograms using CWT, which are passed through CNN for training and class predictions. Alongside CNN, the scalograms produced by CWT are used to find prototypes and criticisms. Once the prototypes and criticism are identified, mean squared error (MSE) is used for comparison to understand the model's decision. At the HMI level, the outcome of this process is then presented for human operator understanding, which increases the trustworthiness of the classification decisions. Furthermore, the mean MSE scores between prototypes and criticisms is used to determine the best voices to be chosen for CWT. When most test samples lie closer to the prototypes than the criticisms, this indicates that the test samples share strong feature similarity with the prototypes, demonstrating that the model has effectively learned the necessary features to make accurate predictions. However, when the criticisms are closer to majority of the test samples than the prototypes, it suggests that the dataset contains large amounts of outlier images, indicating that the model may struggle to generalize and increase the likelihood of misclassification. Therefore, when the mean MSE scores between prototypes and criticisms are compared for each CWT voice, the suitable CWT voices for best model performance happen when the mean MSE prototype score is less than the mean MSE criticisms score. Classification accuracy varies across CWT voices because increasing CWT voices capture more subtle features, thus in this work example-based XAI is used to enhance the classification accuracy by identifying the best CWT voices.

A. Data Preprocessing

The feature preprocessing technique for GOOSE messages from [24] was applied and is summarized in Fig. 2. Every time a GOOSE packet is received, it is broken down into a set of F_{tot} features. In this work $F_{tot} = 26$ features were used, of which the core feature types are described in Table II. An instance of these new received features is then saved into the lists of the values for each feature. To generate a test case a window is slid across the list of features and respective entries for each feature are assembled into a single row vector that preserves the temporal sequence of the data. This method can both be applied in real-time, by keeping the window at the top of the list and each time a new packet arrives a new test case is generated, and post-hoc, where the window can be moved as desired. In this work, the window was selected to contain 96 GOOSE packets, as this was determined experimentally to yield enough data to generate effective scalograms using the CWT and result in acceptable accuracy. The scenario during the experiments was tracked and using the time information contained within each packet each entry in the features lists were labelled using time information and the most critical label. As for the test case generation, the label was selected to be the most critical label of the feature list entries used to generate the test case. That is, if every packet is labelled as "Normal" in a test case but the most recent one is "Attack, DoS," then the label of the test case is "Attack, DoS."

TABLE III
COMPARISON BETWEEN PRE-TRAINED LIGHTWEIGHT MODELS

Network	Accuracy (%)	Training Time (s)
Alexnet	**98.06**	**73.8**
Squeezenet	93.25	83
Efficentnetb0	92.87	287
Googlenet	95.37	210
Inceptionv3	97.32	370
Shufflenet	97.49	91
Resnet50	98.00	215
Nasnetmobile	97.94	498

B. Signal Processing for Feature Extraction

Following the preprocessing stage, the data is transformed using the CWT with the Morse wavelet $h(t)$, where, $x(t)$ is the input signal, a is the scaling parameter and τ is the shifting factor. Morse wavelets are useful when analyzing transient signals where the frequency and amplitude vary over time [25].

$$\mathrm{CWT}\ (\tau, \mathrm{a}) = \frac{1}{\sqrt{\mathrm{a}}} \int_{-\infty}^{\infty} \mathrm{x}\,(\mathrm{t})\,\mathrm{h}^{*}\left(\frac{\mathrm{t}-\tau}{\mathrm{a}}\right) \mathrm{dt} \qquad (1)$$

The number of voices per octave, N, determines how many intermediate scales, j, are placed between octaves, controlling the frequency content. Each scaled wavelet, a_j, is applied across the signal to produce the scalogram using $a_j = (2^{1/N})^{j}$

C. Deep Learning

The generated scalograms are fed to CNN for both training and prediction. The data is split 70/30 for training and testing of the CNN model. This split ensures most of the datasets are used for learning, while the test set remains large enough to cover unseen cases. AlexNet was selected for this study due to its moderate model size, which enables faster training and reduced computational complexity. This was evaluated in this work experimentally against other lightweight models and the results listed in Table III reveal that AlexNet provides the best classification accuracy with the lowest computational time.

The structure of AlexNet is composed of four main layers as depicted in Fig. 3. The input layer takes $227 \times 227 \times 3$ images and turns them into 2D grid of scalogram pixels. The feature extraction layer includes convolution, ReLU, normalization, and max pooling layers. The convolution layer is responsible for

Authorized licensed use limited to: North South University. Downloaded on October 30,2026 at 17:12:18 UTC from IEEE Xplore. Restrictions apply.

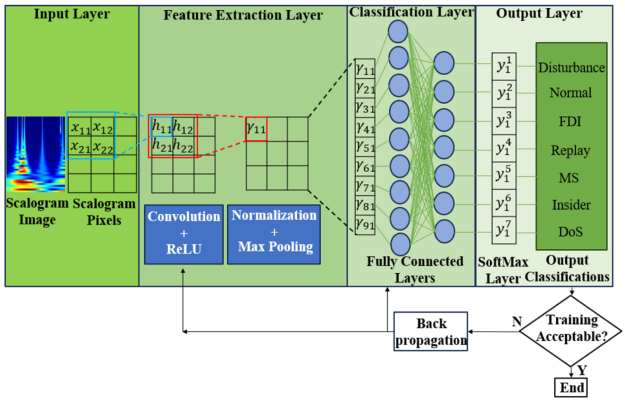


Fig. 3. Overview of CNN model for attack and disturbance classification.

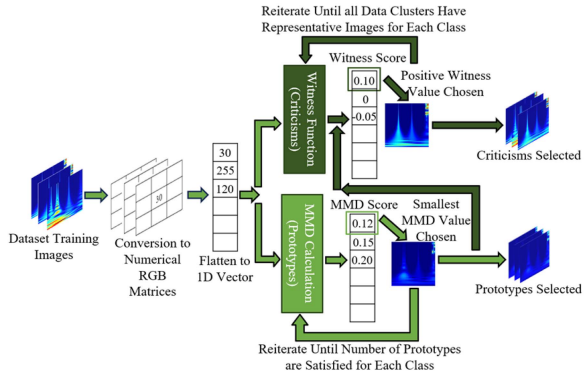


Fig. 4. Implementation of MMD-Critic to enhance interpretability of CNN.

detecting features from the input image using filters. The output feature map, $h_{i,j}$, is calculated in (2) from the dot product between the filter of weights w and the patches of each of the scalogram pixels x, where s represents the height and width of the filter, k and p represent the row and column positions within the convolution kernel of size $s \times s$ and i and j represents the row and column positions of image pixels x.

$$\mathrm{h_{i,j}} = \sum_{\mathrm{k=1}}^{\mathrm{s}} \sum_{\mathrm{p=1}}^{\mathrm{s}} \mathrm{w_{k,l}\, x_{i+k-1,j+p-1}} \tag{2}$$

The rectified linear activation function $ReLU(h)$ and h the output feature map is produced from the convolution layer.

$$\mathrm{ReLU}\ (\mathrm{h}) = \max\{0, \mathrm{h}\} \tag{3}$$

The pooling layer takes the normalized feature output map and merges nearby related features into a single representation. The coefficient γ is calculated using (4), where, G and H are the height and width of the pooling window.

$$\gamma_{i,j} = \max_{\substack{1 \le k \le G \\ 1 \le p \le H}} \{x_{i+k-1,j+p-1}\} \tag{4}$$

The classification layer takes the pooled values from the previous layer and flattens it into a 1D vector. The main component in this layer is the fully connected (FC) layer and the output of each neuron, u_v^l, in the classification layer can be calculated using (5) and each of the neurons after passing the ReLU inside the FC, y_v^l, is calculated as (5), where, l is the layer index, x is the previous layer output, q is the previous layer neuron index and v is the current layer neuron index.

$$u_v^l = \sum_{q=1}^{N} w_{qv}^l\, x_q^l + bias \quad \text{and} \quad y_v^l = f\left(u_v^l\right) \tag{5}$$

The output layer consists of a SoftMax layer, which is responsible for converting real numbers to probability distributions. These probabilities are calculated using (6) where y_z^o has the output probability for a specific class o for the z_{th} sample, $\mathrm{e}^{u_z^o}$ represents the logits of the class being evaluated, $\mathrm{e}^{u_i^o}$ represents logits for each class i, where $i = 1, 2, \ldots, c$ and c is the total number of classes.

$$y_z^o = \frac{\mathrm{e}^{u_z^o}}{\sum_{i=1}^{c} \mathrm{e}^{u_i^o}} \tag{6}$$

Once the SoftMax layer is calculated and the class is predicted, back propagation is done to adjust the weights to achieve the correct classification. The old weights w_{ij}^l are updated using $w_{ij}^l = w_{ij}^l - \Delta w_{ij}^l \eta$ with a learning rate η and gradient of loss with respect to the weight Δw_{ij}^l, where i and j represent the neuron weight index.

D. Example-Based XAI

Example-based XAI is implemented in this work using MMD-Critic, which selects prototype example scalograms obtained from CWT from each class in the dataset that best represents the data distribution that CNN treats as typical for each class. This is then followed by the selection of criticism example scalograms where data distribution diverges from the rest and with variations the model can struggle to capture. Rather than choosing just one prototype, three prototypes were chosen to give a more complete and interpretable explanation thus reducing the risk of misleading conclusions. Fig. 4 depicts the flow of how MMD-Critic functions including the three main components, a kernel, prototypes and criticisms.

1) MMD-Critic Kernel: The kernel serves as the foundational component required to perform the remaining steps in MMD-Critic. The kernel is a function that estimates the similarity between two data points in space. The kernel used was the radial basis function (RBF), where if two points are infinitely apart, it will be 0 and when the two points are equal it will be 1. The RBF kernel was selected because it is widely used for constructing similarity matrices and performs well in clustering applications [26], which is ultimately useful for prototypes and criticism selection. The kernel, k,is calculated using (7) and the scaling parameter δ is calculated using $\delta = \frac{1}{2\sigma^2}$, where, $||x - x'||^2$ is the Euclidean distance (ED) between two points, and σ is the ED of the scalogram pixels between the data in the dataset.

$$k\ (x, x') = \exp\left(-\delta\, ||x - x'||^2\right) \tag{7}$$

The scaling parameter may influence the selection of prototypes and criticisms. Thus, a sensitivity analysis was performed, and the results are given in Table IV. This analysis was done on a disturbance test image. The results reveal that for $\delta > 30$, all three prototypes have very similar MSE values, indicating the prototypes being selected are very similar and fail to capture the

Authorized licensed use limited to: North South University. Downloaded on October 30,2026 at 17:12:18 UTC from IEEE Xplore. Restrictions apply.

TABLE IV
SENSITIVITY ANALYSIS OF δ ON DISTURBANCE TEST IMAGE

δ	Closest Prototype	MSE Value	2nd Closest Prototype	MSE Value	3rd Closest Prototype	MSE Value	Closest Criticsm	MSE Value
1	Disturbance	277	Disturbance	293	Disturbance	316	Disturbance	521
5	Disturbance	104	Disturbance	118	Disturbance	223	Disturbance	277
10	Disturbance	104	Disturbance	142	Disturbance	214	Disturbance	437
30	Disturbance	98	Disturbance	130	Disturbance	250	Disturbance	354
50	Disturbance	99	Disturbance	115	Disturbance	131	Disturbance	277
100	Disturbance	116	Disturbance	120	Disturbance	140	Disturbance	391
500	Disturbance	120	Disturbance	124	Disturbance	140	Disturbance	291

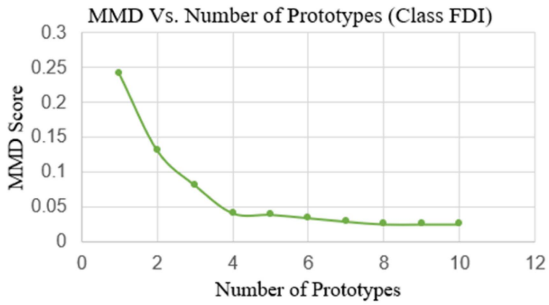


Fig. 5. Relationship between MMD and the number of prototypes.

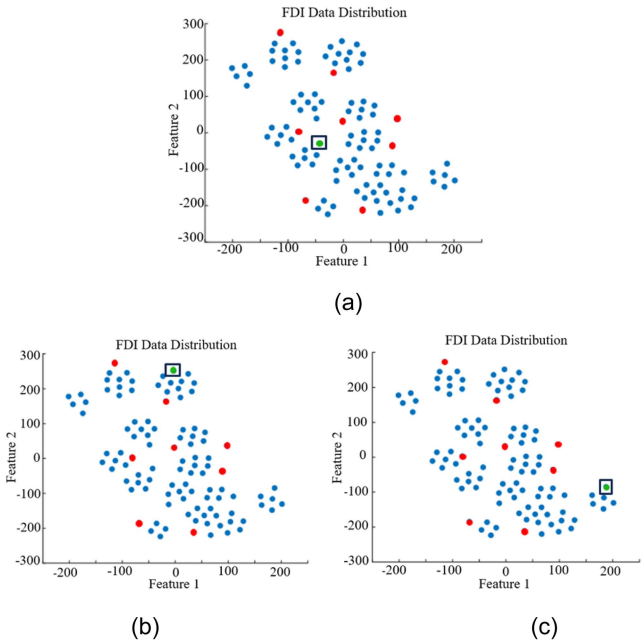


Fig. 6. Criticism selection with witness function (a) witness(x) = 0.0095, (b) witness(x) = -0.0520, (c) witness(x) = 0.0562.

full range of possible clusters in the data. On the other hand, for $\delta < 10$ the prototype MSE values are larger than those of prototype MSE values for the ideal δ value and when $\delta = 5$, the closest prototype and second closest prototype are very close together, indicating full range was not captured. Therefore, this work recommends δ value to be between 10 and 30.

2) MMD-Critic Prototype: With the kernel established, prototypes can then be chosen. Maximum mean discrepancy (MMD) is a measurement that explains how different the two distributions are. The purpose is to evaluate how different the prototype distribution is to the rest of the distributions. MMD is computed using (8) where k is the kernel, m is the number of prototypes, z is the prototypes selected from data points x, and n is the number of data points.

$$MMD^2 = \frac{1}{m^2}\sum_{i,j=1}^{m} k(z_i, z_j) - \frac{2}{mn}\sum_{i,j=1}^{m,n} k(z_i, x_j) + \frac{1}{n^2}\sum_{i,j=1}^{n} k(x_i, x_j) \tag{8}$$

The first term in (8) represents the average proximity between a candidate prototype and the prototypes that have been selected in prior iterations. The second term represents the average proximity between the candidate prototype and all other data, and the third term is the average proximity between all data in the dataset. The goal is to get MMD as close to zero as possible when choosing prototypes, this indicates the prototype represents the data distribution well but are also distinct from each other. In this study, prototypes are chosen by plotting the MMD value to identify the point at which adding additional prototypes results in minimal change to the MMD score. Fig. 5 reveals that around 8 prototypes the curve flattens; hence 8 prototypes were chosen for the FDI class. The MMD score is analyzed after the calculations are completed across all data points, after which the candidate prototype with the lowest MMD score is selected. The process is then repeated with the next candidate prototypes that produce the lowest MMD score alongside the previously chosen prototypes. This iteration process continues until the number of prototypes is met. The MMD scores shown in Fig. 5 are the scores that were produced by the top prototypes that were selected after the MMD score was calculated for all data points after each iteration.

3) MMD-Critic Criticism: Once the prototype examples are selected, the criticism instances can be chosen. Criticisms data distributions are often outliers and represent cases that can be misclassified. The witness function is calculated using (9), where the first term represents the average proximity of the entire data near point x and the second term represents the average proximity of the prototypes near point x.

$$witness\ (x) = \frac{1}{n}\sum_{i=1}^{n} k(x, x_i) - \frac{1}{m}\sum_{j=1}^{m} k(x, z_j) \tag{9}$$

This function provides the means of evaluating the distribution that fits point x better than the prototypes. If *witness* (x) is equal to 0, the distribution of prototypes resembles the distribution of the data as in Fig. 6(a). However, if *witness*(x) is negative, the prototype distribution overestimates the data points, indicating there are only a few data points near x, as in Fig. 6(b). Lastly, if *witness*(x) is positive, the prototype distribution underestimates the data points, indicating there are many data points near x but no prototypes nearby, observed in Fig. 6(c). The criticism is chosen based on the most positive values until all distributions in the dataset have a representative example.

4) MMD-Critic Similarity Comparison: MSE is used to compare how similar the test scalogram is to the prototype and criticisms to understand why the model made its decision. MSE was chosen to be used for comparison since it can provide an easy

Authorized licensed use limited to: North South University. Downloaded on October 30,2026 at 17:12:18 UTC from IEEE Xplore. Restrictions apply.

TABLE V
SIMILARITY COMPARISON METRICS TO A DISTURBANCE CLASS

Similarity Comparison Metric	Closest Prototype Class	Closest Prototype Similarity Value	Closest Criticism Class	Closest Criticism Similarity Value
MSE	Disturbance	111.3	FDI	407.3
Kernel	Disturbance	0.975	FDI	0.925
Cosine Similarity	Disturbance	0.005	Disturbance	0.018

TABLE VI
DATA NORMALIZATION TECHNIQUES COMPARISON

Normalization Technique	Closest Prototype Class	Closest Prototype Similarity Value	Closest Criticism Class	Closest Criticism Similarity Value
Max-Min	Disturbance	0.0017	Disturbance	0.0066
Channel-Wise	Disturbance	0.0122	Disturbance	0.0362
Z-Score	Disturbance	0.0184	Disturbance	0.0681
Unnormalized	Disturbance	98	Disturbance	354

and quick computational measure for comparison [27], which is needed in MMD critic method. This facilitates fast comparison between the prototypes/criticisms and test images, making it well suited for this work. Other similarity metrics are compared in Table V. Cosine similarity was also evaluated; however, cosine similarity works best with feature embeddings rather than pixels [28]. Kernel was also considered, however, the similarity value can be misleading as seen in Table V, where the prototype and criticism values are closely matched. The similarity value using MSE is noticeably different for the same image, but according to the kernel similarity value, it gives the impression that the image is nearly identical to both prototypes and criticism. Thus, MSE was chosen in this study and can be calculated using (10), where, H is the height of the image, W is the width of the image, $f(x, y)$ represents the test image being compared with $\hat{f}(x, y)$, where $\hat{f}(x, y)$ represents the prototypes or criticisms. The lower the MSE, the more similarities the two images share, and the larger the MSE, the more differences the images have.

$$MSE = \frac{1}{HW} \sum_{x=0}^{H-1} \sum_{y=0}^{W-1} \left[f(x, y) - \hat{f}(x, y) \right]^2 \tag{10}$$

5) Data Normalization Techniques: Different data normalization techniques were analyzed to study their effect on the MSE. The first technique is max-min normalization, which rescales the image pixels between the 0 and 1 using (11), where $p_{\min}$ is the minimum pixel value, $p_{\max}$ is the maximum pixel value in the entire data, p is the original pixel value of the image and p_{norm} is the normalized value.

$$p_{norm} = \frac{p - p_{\min}}{p_{\max} - p_{\min}} \tag{11}$$

The second technique is channel-wise normalization, and the third technique is Z-Score normalization, which is calculated using (12), where M is the mean of the image pixels and ζ is the standard deviation of the image pixels.

$$Z\text{-}Score = \frac{p - \mathrm{M}}{\zeta} \tag{12}$$

Table VI shows that the MSE scores are relatively small when normalized and the differences are not significant. Therefore, since the MSE scores are relatively small when normalized, it indicates that the prototypes and criticisms are nearly identical, hindering the explanations. MSE scores without normalization give more meaningful interpretation of how close the prototypes and criticisms truly are and does not further increase computational time.

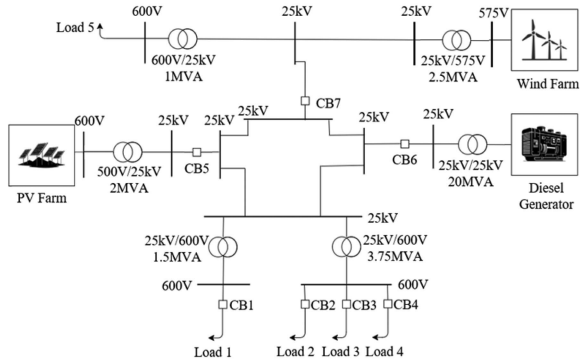


Fig. 7. Single line diagram of the microgrid test system.

TABLE VII
DESCRIPTION OF THE IEDS IN THE TEST SYSTEM

IED Name	Purpose	Actions
CB1	Overcurrent protection for load 1	Open CBs to isolate load 1
CB2	Overcurrent protection for load 2	Open CBs to isolate load 2
CB3	Overcurrent protection for load 3	Open CBs to isolate load 3
CB4	Overcurrent protection for load 4	Open CBs to isolate load 4
CB5	Overcurrent protection for PV farm & 25kV bus	Open CBs to isolate PV farm and 25kV bus
CB6	Overcurrent protection for DG and 25kV bus	Open CBs to isolate DG and 25kV bus
CB7	Overcurrent protection for protection of load 5, wind farm and 25kV bus	Open CBs to isolate load 5, wind farm and 25kV bus
CB_CTRL	Subscribe to GOOSE and control all CBs using SIPROTEC 7SJ64	Close/open the CBs from the input GOOSE messages

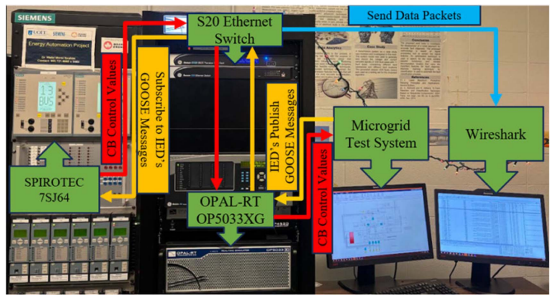


Fig. 8. The experimental setup with HIL.

III. EVALUATION

A. Test System Description and Real-Time HIL

Fig. 7 depicts the single line diagram (SLD) of the microgrid test system, which consists of a 2 MW wind farm, 1.2 MW solar farm, diesel generator, five loads and seven circuit breakers (CBs). Table VII lists the IEDs and their roles for protection of the system. In order to generate real IEC61850 GOOSE message using OPAL-RT for attacks and disturbances cases, a Hardware-in-the Loop (HIL) testing is shown in Fig. 8, which includes Siemens SIPIROTEC-7SJ64 IED, OPAL-RT and Wireshark. An

Authorized licensed use limited to: North South University. Downloaded on October 30,2026 at 17:12:18 UTC from IEEE Xplore. Restrictions apply.

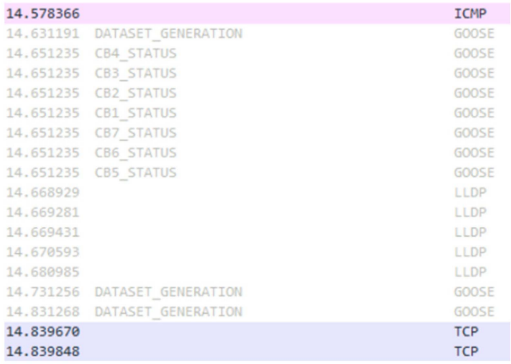


Fig. 9. Collected network traffic from Wireshark showing the GOOSE packets as well as the background traffic.

TABLE VIII
TEST CASE SCENARIOS IN THE TEST SYSTEM

Classification	Number of Scenarios	Number of Cases	Description
Normal	1	903	Operation of system without faults
Disturbance	22	140	3-phase, 2-phase and 1-phase faults
FDI	7	143	FDI attacks launched from IED to falsely trip breakers disrupting normal operation
DoS	7	140	Flooding the network with GOOSE messages, overwhelming the IEDs causing performance to slow down
Replay	7	144	Previous valid faults replayed to falsely cause circuit breakers to open or close, disrupting power flow
Insider	4	144	Associate within the organization falsely opening or closing circuit breakers
MS	7	144	GOOSE messages sent from IEDs are suppressed, not allowing circuit breaker to receive, causing possible damage

OP5033XG OPAL-RT is connected to the PC system that runs RT-LAB software. Wireshark is used to examine network traffic at the packet level and filter the GOOSE message data packets published by the system. Alongside the GOOSE message data packets, additional random background traffic, including TCP, LLDP and ICMP packets were also present and can be seen in Fig. 9. This added noise into the timing of the received GOOSE packets from the network. All communications were passed through a Reason S20 Ethernet Switch. The SIPROTEC 7SJ64 is integrated into the test system, receiving and publishing real GOOSE messages directly to the network with minimal latency. The IED substation configuration language (SCL) file, the masking I/O, and continuous function chart (CFC) logic were configured to work with the simulation model in this study.

Table VIII lists the cases studied in this work, including normal operation, disturbances, and various attacks such as FDI, DoS, replay, insider, and MS. The number of scenarios represents the different situations. For instance, in the context of disturbances, one scenario may involve a three-phase grounded fault, whereas another may involve a three-phase ungrounded fault. The number of cases denotes the specific scenarios recorded within each category, resulting in 1618 total cases in the dataset. The data packets of the GOOSE messages captured in Wireshark contain cyber features that are used for the scalograms, which were described in Section II.

TABLE IX
CLASS-WISE PERFORMANCE RESULTS

Class	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
Normal	99.42	98.87	100.00	99.43
Disturbance	99.23	100.00	90.47	95.00
FDI	99.04	100.00	88.89	94.11
DoS	99.81	100	97.56	98.76
Replay	98.84	87.5	100.00	93.33
Insider	99.81	97.67	100.00	98.82
MS	100.00	95.56	100.00	97.72
Averaged	**99.45**	**97.08**	**96.70**	**96.73**

B. Performance Evaluation Metrics

The overall classification accuracy, τ, is calculated using (13), which measures the ability of the model to classify each case including each individual attack, where N_c is the number of classes, N_o is the total number of test cases, TP_c is the correct predictions made for that class, and c represents a class.

$$\tau = \frac{\sum_{c=1}^{N_c} TP_c}{N_o} \tag{13}$$

To assess the performance for each class, the per-class precision, recall, F1-score, and accuracy is calculated. Precision, p_c, measures of how many of the predictions made by the model labeled as positives were correct and is given by (14), where, true positive (TP_c) refers to the number of samples correctly identified to belong to the positive class, and false positive (FP_c) occurs when the model incorrectly predicts a sample as the target class. Recall, r_c, measures how many of the actual positive cases the model successfully detected and is calculated using (14), where, false negative (FN_c) occurs when the model predicts that a sample is not the target class.

$$p_c = \frac{TP_c}{TP_c + FP_c} \quad \text{and} \quad r_c = \frac{TP_c}{TP_c + FN_c} \tag{14}$$

The F1-Score, $F1_c$, is calculated using both precision and recall while the Classification Accuracy Per Class, a_c, measures the classification accuracy for each class individually using (15).

$$F1_c = 2 \times \frac{(p_c * r_c)}{p_c + r_c} \quad \text{and} \quad a_c = \frac{TP_c + TN_c}{N_o} \tag{15}$$

C. Performance Evaluation Results Without MMD-Critic

Table IX lists the results showing an overall classification accuracy of 98.06%. Key hyperparameters used in this study are epochs, learning rate and batch size. The number of epochs chosen was 5 to ensure the model receives sufficient training to generalize effectively, while also optimizing training time. The batch size was selected as 20 to maintain frequent weight updates without compromising computational efficiency. Lastly, the learning rate used was 0.0001 to ensure that CNN training loss converges whilst avoiding slow training. Fig. 10 illustrates CNN training process providing insight into the model convergence and generalization behavior. Toward the end of training, the model beings to converge without indications of overfitting.

Authorized licensed use limited to: North South University. Downloaded on October 30,2026 at 17:12:18 UTC from IEEE Xplore. Restrictions apply.

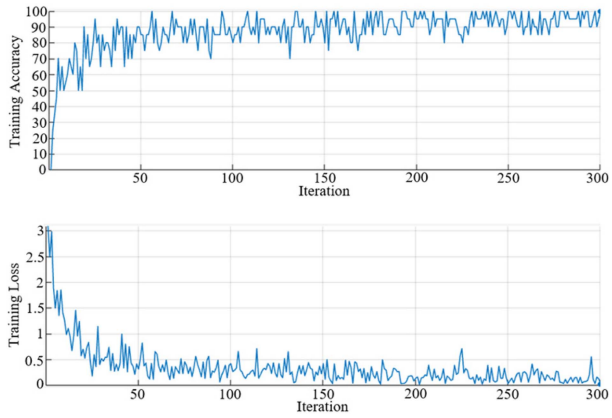


Fig. 10. The accuracy and loss of model.

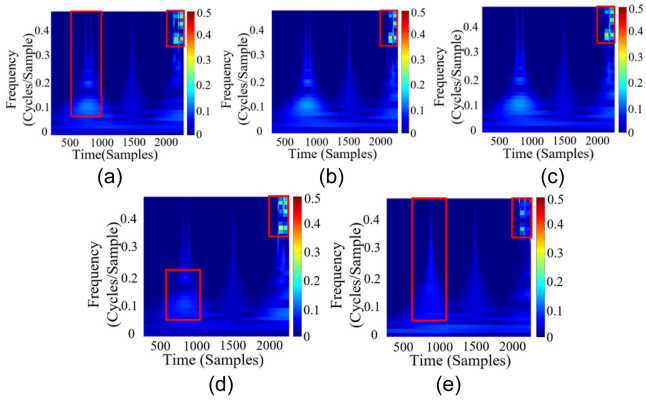


Fig. 11. MMD-Critic scalograms of a correct classification case. (a) Input image, (b) closest prototype, (c) second closest prototype, (d) third closest prototype, (e) closest criticism.

D. Interpretability of Results Using XAI

Fig. 11(a) represents the input scalogram to be compared to the prototypes and criticism. Fig. 11(b) represents the closest prototype, with only minor differences observed in the upper region toward the end of the scalogram. Fig. 11(c) represents the second closest prototype, with slightly more differences observed within the same region of the scalogram. Fig. 11(d) is the third closest prototype, with more evident differences across the scalogram compared to the previous prototypes. Fig. 11(e) represents the closet criticism, with major differences across the scalograms. The three prototypes closest to the input scalogram were of the disturbance class. Visually comparing the input image to the prototypes and the criticism, the prototypes show stronger similarity. This gives more trustworthiness in the model's decision since the features of the input image are like that of the typical disturbance class scalograms. The criticism was FDI, indicating that input scalogram has similar features to the scalogram in the FDI class that are underrepresented but there are still key differences between the two. MSE is used to understand which prototypes and criticisms are closer to each input scalogram, when the input is much closer to prototypes than criticism. The criticism contributes far less when the MSE score is much higher and has concrete visual

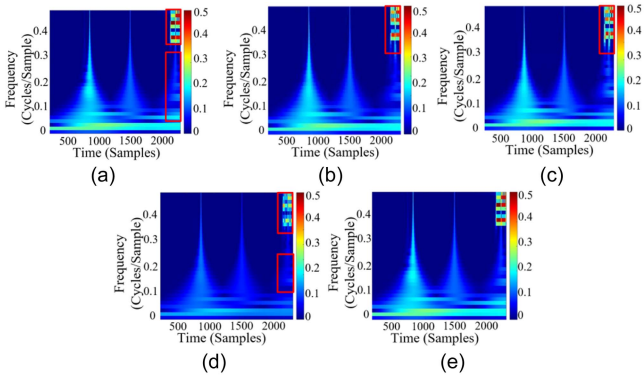


Fig. 12. MMD-Critic scalogram of a misclassification case. (a) Misclassified input image, (b) closest prototype, (c) second closest prototype, (d) third closest prototype, (e) closest criticism.

differences compared to the prototypes, showing that while they share some similarity, they are not close enough to push the model toward misclassification. As a result, the prediction is disturbance remains confident and highly likely to be correct. Fig. 12 shows another case, where Fig. 12(a) represents a misclassified input scalogram to be compared to the prototypes and criticism. Fig. 12(b) represents the closest prototype, with only slight differences observed in the highlighted area of the scalogram. Fig. 12(c) represents the second closest prototype, which closely resembles the first prototype but with a few differences visible in the highlighted region of the scalogram. Fig. 12(d) is the third closest prototype showing more noticeable differences near the end of the scalograms in both the upper and lower regions. Fig. 12(e) represents the closest criticism, which is seen to be identical to the input scalogram. In this case, the criticism is visually much more like the input image than the prototypes, effectively showing that the model's prediction may be untrustworthy. For this case, the prototypes closest to this scalogram were all the replay class and the closest criticism was of the FDI class. However, the model predicted the replay class, but the true classification was FDI class. Since the criticism and input scalogram are practically identical, CNN could not learn the features well enough to classify it correctly. Also, since the closest prototypes were of the replay class, it indicates that this image is near the decision boundary between the replay and FDI class, creating more uncertainty for CNN. This indicates how MMD-Critic can determine trustworthiness and identify possible misclassifications. Also, the computational time to find the prototypes and criticism for each class is on average 3.58 seconds. As finding these only require the training data, having them computed in advance and readily available is the most effective approach for real-time microgrid IDS application.

E. Model Performance Enhancement

1) Mean MSE for Voice Selection: Fig. 13 shows the relationship between mean MSE of prototypes and criticism. When the difference between mean MSE of prototypes and mean MSE of criticisms is negative, the classification is the highest, and it begins to drop when the difference between MSE of criticism and MSE of prototypes is positive. Scalograms generated

Authorized licensed use limited to: North South University. Downloaded on October 30,2026 at 17:12:18 UTC from IEEE Xplore. Restrictions apply.

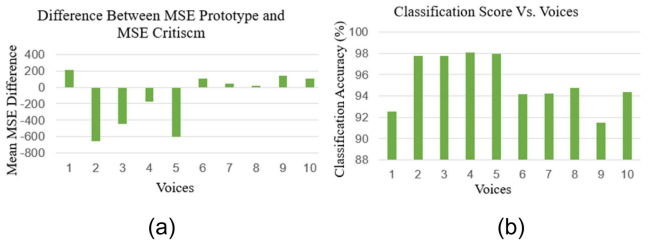


Fig. 13. A comparison between the difference in mean MSE and classification accuracies of CWT voices from 1-10 (a) difference in mean MSE between criticism and prototypes, (b) classification accuracies.

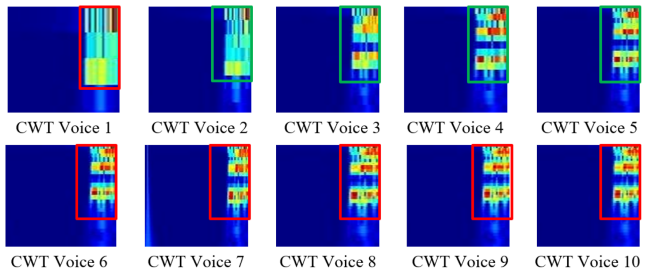


Fig. 14. CWT scalogram features information for voices 1-10.

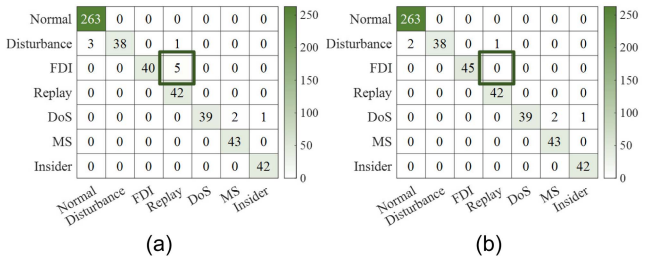


Fig. 15. Confusion matrix (a) without MMD-Critic Criticisms, with 5 FDI cases incorrectly classified as replay attacks, and confusion matrix (b) following the use of Criticisms, with correct classification of FDI cases.

with CWT voices ranging from 1 to 10 show that increasing voices initially improves time-frequency resolution, revealing important patterns as in Fig. 14. For this dataset, the range between voices 2-5 was the best to use. In the first voice, the scalogram does not have enough frequency content produced by the CWT and is low-resolution as observed in the highlighted region of the scalogram. This ultimately makes it difficult for CNN to distinguish from the other classes. As voices increased, more feature information was introduced for CNN to distinguish between classes, and the more refined these scalograms became, as seen in the region of interest in the scalograms from voices 2-5. However, after voice 5, the classification begins to drop, and this is due to the tradeoff of redundancy where additional voices add more subtle feature information, which confuses CNN, leading to misclassification.

2) CNN Classification Improvement Using Criticisms: To further improve the classification, criticisms can identify precisely which test case was misclassified and why. Fig. 15(a) shows the confusion matrix of where the model struggles to classify test cases and what the CNN misclassified it as. Test cases where the scalogram is identical to a criticism can make it hard for CNN to learn the features of these scalograms due to limited training images with similar features.

To enhance the performance of CNN, diversifying the training dataset by adding additional criticism images that MSE score showed to be identical to the criticisms into the training data and retraining CNN is required. The criticisms images that were added were generated using a higher CWT voice of 10. This will improve the coverage of the feature space that enables CNN to learn the features required for correct classification. The criticisms from MMD-Critic identified 5 FDI cases, as seen in Fig. 12, were all identical to the same FDI criticism, and duplicates of this criticism were added back into the training data for re-training. The objective is to incorporate known outlier images back into the training dataset to improve class representation to reduce misclassification. The model initially trained without the additional criticisms in Fig. 15(a) shows 5 FDI cases misclassified as replay. After re-training, it can be seen in (b) that the misclassified cases that were identical to the criticism are now classified in their appropriate classes. The use of MMD-Critic can determine the specific misclassified cases that can be successfully classified correctly through feedback of criticisms and re-training the model. The other misclassified cases that were not identical to the criticisms but have greater feature similarity to the criticisms than the prototypes could not be classified correctly through re-training and without MMD-Critic, it would not be possible to identify these cases. This shows that MMD-Critic can improve the performance of CWT-CNN by finding certain cases that are outliers, which can be re-trained and classified correctly. To ensure variability during retraining, the random number generator was used multiple times to evaluate the model at different seed values and the results continued to show FDI cases to be correctly classified.

TABLE X
ACCURACY COMPARISON WITH BASELINE TECHNIQUES

Methodology	Classification Accuracy (%)
Linear	76.87
SVM + CNN	91.09
SVM	91.32
KNN	92.19
Kernel	95.98
CWT + CNN	98.06
This Work (CWT-CNN-XAI)	**98.85**

3) Comparison With Baseline Machine Learning Models: The proposed methodology is compared to baseline machine learning models in Table X. These include machine learning classifiers, implemented using Error Correcting Output Codes (ECOC) to achieve multi-class classification, and the base CWT with a CNN, without the MMD-Critic-based feedback implemented. As is seen in Table X, these classifiers are unable to achieve as high a detection accuracy as the proposed methodology, which achieves 98.85%. While CWT and CNN can achieve 98.06%, they cannot distinguish between the difficult cases without the feedback from XAI as shown earlier.

4) Comparison With Other Studies in Literature: The works in [29], [30], [31], [32], [33] propose various learning-based

Authorized licensed use limited to: North South University. Downloaded on October 30,2026 at 17:12:18 UTC from IEEE Xplore. Restrictions apply.

TABLE XI
COMPARISON OF MICROGRID ATTACK DETECTION METHODS IN LITERATURE

Study	Methodology	Detection Accuracy (%)
[29]	WT-SVD-SVM	97.34
[30]	HHT-DL	98.09
[31]	LSTM	64.55
[32]	RNN	97.67
[33]	HHT-LSTM-VAE-Bi-GRU	97.28
This work	**(CWT-CNN-XAI)**	**98.85**

TABLE XII
MMD-CRITIC (TOP ROWS) AND PROTODASH (BOTTOM ROWS) COMPARISON

Prediction	Closest Prototype	MSE Value	2nd Closest Prototype	MSE Value	3rd Closest Prototype	MSE Value	Closest Criticism	MSE Value
Disturbance Replay	Disturbance Replay	74 154	Disturbance Replay	114 205	Disturbance Replay	212 214	FDI FDI	434 0
Disturbance Replay	Disturbance FDI	130 0	Disturbance Replay	271 205	Disturbance Replay	379 214	FDI Insider	535 959

methods to detect and classify attacks on microgrids. These methods were applied to the dataset generated in this work, and the resulting accuracies shown in Table XI. The proposed methodology outperforms the methods in literature in terms of detection accuracy. The improvement in accuracy from using the proposed methodology of feedback from MMD-Critic improves the performance of the classifier, which helps separating the more difficult to detect classes as shown earlier.

5) Comparison of ProtoDash and MMD-Critic: Table XII lists prototypes and criticism chosen for two different predictions by MMD-critic. For first prediction, which was correctly classified as disturbance, both algorithms selected representative prototypes that effectively explained the model's decision. However, the second prediction was misclassified as a replay, while the actual class was FDI. The response from MMD-critic explains that it shares similar features to that of the replay class however, the criticism indicates it shares much more feature similarity to an FDI class, which indicates possible misclassification. ProtoDash's response shows that there are no possible misclassification and the prototypes that share the highest feature similarity is between FDI and replay, leading to further confusion. ProtoDash can be seen to struggle to find meaningful criticisms. Therefore, for meaningful explanations, MMD-critic is the superior example-based XAI method.

6) LIME and SHAP Comparison With MMD-Critic: The explanations of SHAP and LIME can be seen in Fig. 16 for two different disturbance test images with similar features. The features identified as important by two XAI models vary across the two images, even though the features are very similar between the cases. Therefore, without the ability to compare across multiple explanations, it is hard to determine which explanation would be trustworthy for the decision of the model. Example-based XAI can overcome the limitation of LIME and SHAP of trustworthiness by comparing the test case between multiple prototypes and criticisms. A trustworthy and correct explanation is when prototypes are similar in features and are of the same class or if the closest prototype is identical. This is demonstrated in Fig. 16(a) and (b) where the closest and second closest prototypes are shown. It is evident that in both cases,

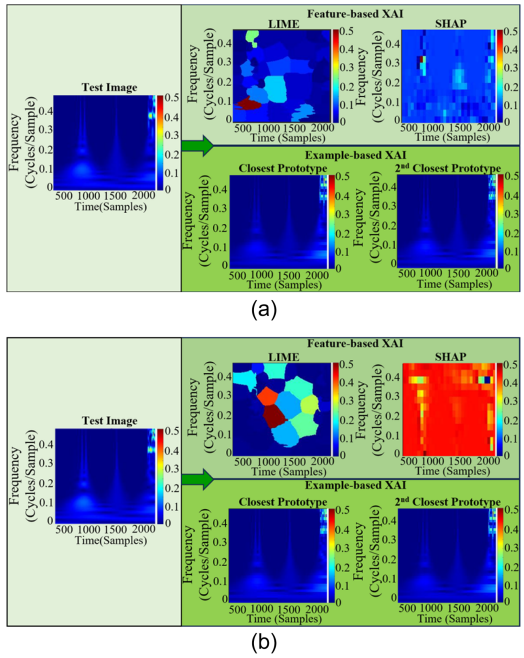


Fig. 16. Model explanations between LIME, SHAP and MMD-Critic. (a) First disturbance case, (b) second disturbance case.

the prototypes share strong visual similarity to one another, closely resemble their corresponding test case and classified as disturbance. Furthermore, it can also be observed that the closest prototype for the original test case was identified as the second-closest prototype to the other test case, indicating a trustworthy and consistent explanation. A less trustworthy decision can occur if criticisms have more similarity than prototypes or if prototypes are of different classes. Also, LIME and SHAP are known to assume feature independence, which breaks natural correlations in time-frequency representations and further reduces trustworthiness of the methods [34]. Lastly, MMD-Critic explanations are much more straightforward for humans to understand, further demonstrating its feasibility.

7) Large-Scale Microgrid Systems: To further validate the proposed method, a large scale microgrid was used form Indigenous Kluane First Nation energy project in [35] for which a dataset consisting of attacks and disturbances was generated through OPAL-RT. The dataset generated consisted of twice the number of samples as the original dataset and the classification accuracy average across CWT voices was 98.25%. The proposed methodology recommended voice selection from mean MSE was between 1 to 3, which produced the highest classification accuracy. Lastly, MMD-Critic criticisms identified 2 disturbance cases and 1 MS case that were identical to the criticisms, which were then added back into the training data with a higher CWT voice and classified correctly through re-training the model.

8) Operator's GUI At HMI Level: A requirement for a successful XAI-based IDS is human-machine interaction at HMI level to allow the operator to see the reasoning behind decisions of models. Fig. 18 shows the GUI, where the input image is shown along with the model's prediction. Followed by three prototype scalograms with the closest classes, along with each

Authorized licensed use limited to: North South University. Downloaded on October 30,2026 at 17:12:18 UTC from IEEE Xplore. Restrictions apply.

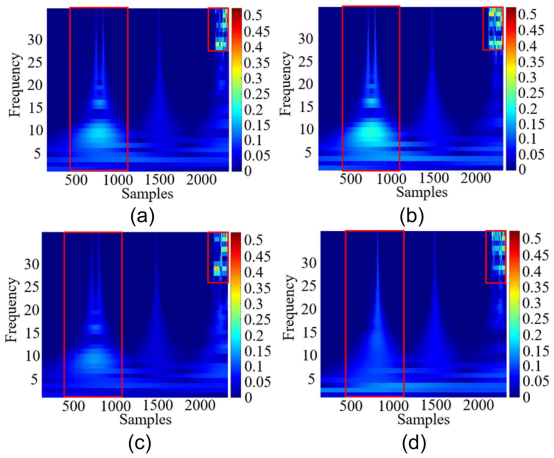


Fig. 17. Prototypes comparison. (a) Closest prototype, (b) second closest prototype, (c) third closest prototype, (d) fourth closest prototype.

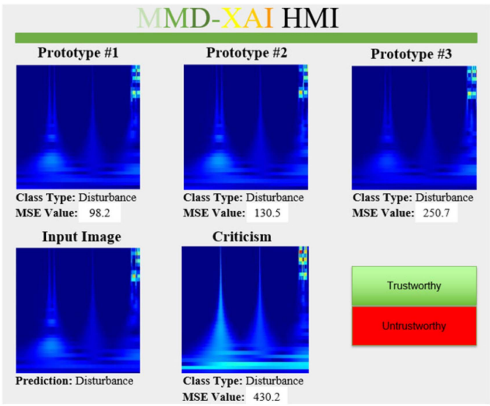


Fig. 18. MATLAB-based user interface using MMD-Critic.

TABLE XIII
CLOSEST PROTOTYPES TO DISTURBANCE TEST IMAGES

Closest Prototype	MSE Value	2nd Closest Prototype	MSE Value	3rd Closest Prototype	MSE Value	4th Closest Prototype	MSE Value
Disturbance	135	Disturbance	154	Disturbance	206	Disturbance	358
Disturbance	58	Disturbance	124	Disturbance	206	Disturbance	308
Disturbance	87	Disturbance	110	Disturbance	199	Disturbance	307
Disturbance	74	Disturbance	114	Disturbance	212	Disturbance	318

of the MSE scores and lastly, the closest criticism scalogram, the class and the MSE score are shown. Displaying three prototypes provides operators with the ability to see similar reoccurring features between the different prototypes and the test image. If the three prototypes all came back as the same class and are similar, it provides more trustworthiness in the prediction of CNN, rather than just having a singular prototype with the same class. Table XIII shows four different prototypes and the MSE values to the different disturbance test images. The ideal number of prototypes is only three since the sudden increase in the MSE value for the fourth prototype indicates increased feature disparity. Fig. 17 demonstrates the feature disparity between the first, second, third and fourth prototypes. It is evident that in the highlighted regions of the spectrograms, the closest prototypes from one to three share similar features and the MSE values are much lower. However, when a fourth prototype is added, a clear distinction can be observed in the spectrograms, where the highlighted areas differ from the other three prototypes. Therefore, the addition of more prototypes can be seen to have visual feature differences with other prototypes and produce a higher MSE score, which can degrade operator comprehension and reduce confidence in the decision. The operators using this interface can quickly compare the incoming input image visually to the prototypes and criticism but also use the MSE score to understand and evaluate the decision. A green box will appear on the HMI depending on if the prototypes MSE values are closer to than the criticism if trustworthy. A red box will appear if criticisms are closer to the prototypes if untrustworthy. MMD-Critic in HMI helps explain model's decisions using examples and through comparison with the input image, as this is the representation most easily understood by humans.

IV. CONCLUSION

In this work, a novel XAI-Based IDS is developed for disturbance and attack detection and classification in Cyber-Physical Microgrids. The proposed approach uses CWT to analyze the time-frequency relationship of the cyber features from GOOSE messages. The decisions made by CNN were interpreted by XAI-example-based methodology MMD-critic. MMD-Critic helps show interpretability by showing similar examples to typical cases, making the decisions understandable and trustworthy. Alongside interpretability, MMD-Critic was used to help determine the number of voices best suited to be used with CWT for the best classification possible. MMD-Critic can also help understand areas of misclassification using criticisms, which are known to be outlier images that the model was not able to learn the important features to correctly classify. To address this, more images like the criticism with a higher CWT voice have been added into the training data, which ultimately have helped the model learn the required features to classify the misclassified cases correctly, increasing classification accuracy to 98.85%. Lastly, an operator interface is developed at the HMI level, aimed at enhancing the interpretability of decisions. These results show that time-frequency images alongside classifiers and interpretability models can provide accurate and informed decisions.

REFERENCES

[1] D. Qiu, M. Liu, R. Zhang, T. Luo, A. Griffo, and X. Zhang, "Cyber-physical real-time digital simulation for cybersecurity analysis in microgrids," *IEEE Trans. Ind. Cyber-Phys. Syst.*, vol. 3, pp. 429–441, May 2025.
[2] A. A. Nassar, M. Oinonen, and W. G. Morsi, "An integrated trustworthy detection and classification of cyber-physical attacks in the presence of disturbances using morphological image processing and explainable AI," *IEEE Trans. Ind. Cyber-Phys. Syst.*, vol. 3, pp. 442–453, Jul. 2025.
[3] W. M. Hamanah, M. I. Hossain, M. Shafiullah, and M. A. Abido, "AC microgrid protection schemes: A comprehensive review," *IEEE Access*, vol. 11, pp. 76842–76868, 2023.
[4] S. Ishikawa et al., "Example-based explainable AI and its application for remote sensing image classification," *Int. J. Appl. Earth Observation Geoinf.*, vol. 118, Apr. 2023, Art. no. 103215.
[5] S. Jadidi, H. Badihi, and Y. Zhang, "Active fault-tolerant and attack-resilient control for a renewable microgrid against power-loss faults and data integrity attacks," *IEEE Trans. Cybern.*, vol. 54, no. 4, pp. 2113–2128, Apr. 2024.

Authorized licensed use limited to: North South University. Downloaded on October 30,2026 at 17:12:18 UTC from IEEE Xplore. Restrictions apply.

[6] S. Wu, D. Ye, and X. Shao, "Event-based distributed resilient control strategy for microgrids subject to disturbances and hybrid attacks," *Appl. Math. Computation*, vol. 459, Dec. 2023, Art. no. 128273.
[7] O. A. Beg, L. V. Nguyen, T. T. Johnson, and A. Davoudi, "Signal temporal logic-based attack detection in DC microgrids," *IEEE Trans. Smart Grid*, vol. 10, no. 4, pp. 3585–3595, Jul. 2019.
[8] S. Sahoo, S. Mishra, J. C.-H. Peng, and T. Dragičević, "A stealth cyber-attack detection strategy for DC microgrids," *IEEE Trans. Power Electron.*, vol. 34, no. 8, pp. 8162–8174, Aug. 2019.
[9] S. Tan, P. Xie, J. Sun, J. C. Vasquez, and J. M. Guerrero, "A novel observer design for disturbance and attack-tolerance state estimation in microgrids," *IEEE Trans. Smart Grid*, vol. 17, no. 2, pp. 1574–1588, Mar. 2026.
[10] A. Andreotti, B. Caiazzo, D. G. Lui, A. Petrillo, and S. Santini, "Enhancing resilience of islanded microgrids under disturbances, delays, and DoS attacks through a novel digital predictor method," *IEEE Trans. Cybern.*, vol. 55, no. 3, pp. 1310–1322, Mar. 2025.
[11] X. Zhang et al., "Event-triggered resilient recovery learning control protocol for interconnected DC microgrids with distributed attack detection," *Sustain. Energy Grids Netw.*, vol. 38, Jun. 2024, Art. no. 101364.
[12] Y. Wan and T. Dragičević, "Data-driven cyber-attack detection of intelligent attacks in islanded DC microgrids," *IEEE Trans. Ind. Electron.*, vol. 70, no. 4, pp. 4293–4299, Apr. 2023.
[13] M. A. S. P. Dayarathne, M. S. M. Jayathilaka, R. M. V. A. Bandara, V. Logeeshan, S. Kumarawadu, and C. Wanigasekara, "Mitigating cyber risks in smart cyber-physical power systems through deep learning and hybrid security Models," *IEEE Access*, vol. 13, pp. 37474–37492, 2025.
[14] P. Arévalo, A. Cano, D. Benavides, and F. Jurado, "Fault analysis in clustered microgrids utilizing SVM-CNN and differential protection," *Appl. Soft Comput.*, vol. 164, Oct. 2024, Art. no. 112031.
[15] Y. Xia, Y. Xu, S. Mondal, and A. K. Gupta, "A transfer learning-based method for cyber-attack tolerance in distributed control of microgrids," *IEEE Trans. Smart Grid*, vol. 15, no. 2, pp. 1258–1270, Mar. 2024.
[16] N. Bhusal, M. Gautam, and M. Benidris, "Detection of cyber-attacks on voltage regulation in distribution systems using machine learning," *IEEE Access*, vol. 9, pp. 40402–40416, 2021.
[17] K. Bitirgen and Ü. B. Filik, "A hybrid deep learning model for discrimination of physical disturbance and cyber-attack detection in smart grid," *Int. J. Crit. Infrastructure Protection*, vol. 40, Mar. 2023, Art. no. 100582.
[18] S. R. Kasimalla et al., "Deep learning-enabled system diagnosis in microgrids: A feature-feedback GAN approach," in *Proc. IEEE Int. Conf. Commun., Contr. Comp. Tech. Smart Grids*, North York, ON, Canada, 2025, pp. 1–6.
[19] A. Hoenig, K. Roy, Y. T. Acquaah, S. Yi, and S. S. Desai, "Explainable AI for cyber-physical systems: Issues and challenges," *IEEE Access*, vol. 12, pp. 73113–73140, 2024.
[20] R. Machlev, M. Perl, J. Belikov, K. Y. Levy, and Y. Levron, "Measuring explainability and trustworthiness of power quality disturbances classifiers using XAI—Explainable artificial intelligence," *IEEE Trans. Ind. Informat.*, vol. 18, no. 8, pp. 5127–5137, Aug. 2022.
[21] M. R. Dehbozorgi, M. Rastegar, and M. F. M. Arani, "False data injection attack detection and localization framework in power distribution systems using a novel ensemble of CNNs and explainable artificial intelligence," *IEEE Trans. Ind. Appl.*, vol. 61, no. 3, pp. 4801–4811, May/Jun. 2025.
[22] R. Upadhyay et al., "XAI-based DoS attack detection framework for reliable energy distribution in smart grid systems," in *Proc. 3rd Int. Conf. Intel. Data Commun. Tech. Internet Things*, Bengaluru, India, 2025, pp. 340–345.
[23] S. Hariharan et al., "XAI for intrusion detection system: Comparing explanations based on global and local scope," *J. Comput. Virol. Hacking Techn.*, vol. 19, no. 2, pp. 217–239, Jun. 2023.
[24] M. Oinonen and W. G. Morsi, "Cyberattack detection and classification using new custom filter design and explainable-deep learning in cyber-physical systems," *IEEE Trans. Ind. Cyber-Phys. Syst.*, vol. 4, pp. 225–237, 2026.
[25] O. A. Allan and W. G. Morsi, "A new passive islanding detection approach using wavelets and deep learning for grid-connected photovoltaic systems," *Electric Power Syst. Res.*, vol. 199, Oct. 2021, Art. no. 107437.
[26] F. Baffour-Awuah, M. Fazli, and D. A. Striegel, "Multi-scalar spectral clustering: A new approach to capture scale-dependent persistent clusterings," *IEEE Access*, vol. 13, pp. 190843–190858, 2025.
[27] Z. Wang, A. C. Bovik, H. R. Sheikh, and E. P. Simoncelli, "Image quality assessment: From error visibility to structural similarity," *IEEE Trans. Image Process.*, vol. 13, no. 4, pp. 600–612, Apr. 2004.
[28] Q. Wang and K. A. Lee, "Cosine scoring with uncertainty for neural speaker embedding," *IEEE Signal Process. Lett.*, vol. 31, pp. 845–849, 2024.
[29] M. Dehghani et al., "Cyber-attack detection in DC microgrids based on deep machine learning and wavelet singular values approach," *Electronics*, vol. 10, no. 16, Aug. 2021, Art. no. 1914.
[30] H. Cui, X. Dong, H. Deng, M. Dehghani, K. Alsubhi, and H. M. A. Aljahdali, "Cyber attack detection process in sensor of DC micro-grids under electric vehicle based on Hilbert–Huang transform and deep learning," *IEEE Sensors J.*, vol. 21, no. 14, pp. 15885–15894, Jul. 2021.
[31] A. S. Mohamed and D. Kundur, "On the use of reinforcement learning for attacking and defending load frequency control," *IEEE Trans. Smart Grid*, vol. 15, no. 3, pp. 3262–3277, May 2024.
[32] M. Karanfil et al., "Detection of microgrid cyberattacks using network and system management," *IEEE Trans. Smart Grid*, vol. 14, no. 3, pp. 2390–2405, May 2023.
[33] Z. Ali et al., "A novel hybrid signal processing based deep learning method for cyber-physical resilient harbor integrated shipboard microgrids," *IEEE Trans. Ind. Appl.*, vol. 62, no. 1, pp. 382–401, Jan./Feb. 2026.
[34] M. Oinonen and W. G. Morsi, "Interpretable intelligent intrusion detection systems for cybersecurity in smart digital substations," *Electric Power Syst. Res.*, vol. 258, Sep. 2026, Art. no. 113087.
[35] J. Zrum et al., "Kluane Ǹ-Ts'i (Wind) project power system impact study–V1.0," Yukon, Mar. 2022. [Online]. Available: https://www.yukonu.ca/sites/default/files/inline-files/IE0110%20-%20Power%20System%20Impact%20Study%20-%20220309.pdf



Gurmaan S. Panesar (Student Member, IEEE) was born in Toronto, Canada in 2001. He received the B.Sc. (Eng.) degree in electrical engineering from Ontario Tech University, Oshawa, ON, Canada, in 2024, where he is currently working toward the M.A.Sc. degree in electrical and computer engineering. From 2022 to 2023, he was an Electrical Engineering Intern, focusing on electrical design and project implementation with Honda of Canada Manufacturing, Alliston, ON, Canada. His research interests include power system protection, machine learning, and cyber security.



Matthew Oinonen (Member, IEEE) was born in Toronto, ON, Canada in 1999. He received the B.Sc. (Eng.) degree in electrical engineering from Ontario Tech University, Oshawa, ON, Canada, in 2022, where he is currently working toward the Ph.D. degree in electrical and computer engineering after transferring directly into the program from the M.A.Sc. degree in 2024. He was an Engineering Student in 2019 and Engineering Student Intern from 2021 to 2022 in electronics research and development with Siemens Canada Limited, Peterborough, ON, Canada. His research interests include signal processing, machine learning, energy monitoring, and cyber security.



Walid G. Morsi (Senior Member, IEEE) was born in Ismailia, Egypt in 1975. He received the B.Sc. (Eng.) and M.Sc. (Eng.) degrees in electrical engineering from Suez Canal University, Ismailia, Egypt, in 1998 and 2002, respectively, and the Ph.D. degree in electrical engineering from Dalhousie University, Halifax, NS, Canada, in 2009. He was a Killam Memorial Pre-doctoral Scholar with Dalhousie University and then an Assistant Professor with the Department of Electrical and Computer Engineering, University of New Brunswick, Fredericton, NB, Canada. He is currently a Professor with the Department of Electrical, Computer and Software Engineering, Faculty of Engineering and Applied Science, Ontario Tech University, Oshawa, ON, Canada. His research interests include smart grid, power quality/disturbance data analytics, cybersecurity, AI application in management and automation of electric energy systems. Dr. Morsi is a registered Professional Engineer (P.Eng.) with the Association of Professional Engineers in Ontario (PEO).

Authorized licensed use limited to: North South University. Downloaded on October 30,2026 at 17:12:18 UTC from IEEE Xplore. Restrictions apply.